

Calibration as Linkage Integrity

Does a knowledge base's rigor survive contact with a reasoner?

Submission to the FLF Epistemic Case Study Competition ("Epistack")

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Materials: a written argument (this document), a runnable instrument (`epistack_eval.py`), and its run outputs across three cases and two models, all in the linked repository.

The core submission

The real test of the rigor of a knowledge base is how well it survives contact with a reasoner. Ideally the reasoner leaning only on the base should be as confident as the evidence in the base warrants. Any more or less confidence is a failure of the base and not a failure of the reader. My submission is a way to measure that deviation, and an instrument that does it on real contested cases.

I am showing this on all three cases of different shape. One is contested, one is settled, and the third is ill-posed. I used the same instrument, with two different models (Sonnet and Opus), to ensure it works as infrastructure. A method that only works on one case is not yet infrastructure.

The problem

The true test of the rigor of a knowledge base lies in the confidence of the reasoner. A base can have a very well organised structure and confident-sounding claims and still leave the reasoner who leans on it believing things the evidence does not support. That gap is invisible when we only inspect the artifact, because the failure shows up somewhere else, in the reasoner who used it.

Why this is the layer that matters

Input-side inspection can miss the effect on a reasoner. Measuring the linkage integrity is the key here. Structure and provenance will continue to be important, and they certify the quality of the inputs. But even with them, the result may still lead to the reasoner double-counting correlated evidence, or missing the one choice the answer actually hinges on. That is why the base has to be tested by what it does to a reasoner, not only by how it looks.

I have put in effort to check and build an infrastructure to ensure that linkages actually hold, from the output side.

Is it beating off-the-shelf deep research? In fact it is not competing with deep research; in terms of layers, it sits one layer above it. Whether the artifact the reasoner tries to lean on is actually rigorous or not is not reflected from the deep research that handed the reasoner the artifact, the set of claims, and the knowledge base. My instrument does exactly that. You can even run deep research to build the base, then point the instrument at it to see where a reasoner ends up more or less confident than the evidence warrants, which claims are load-bearing, and where correlated evidence gets double-counted. So deep research can produce the artifact; my instrument measures whether the artifact survives contact with the reasoner. To finish the loop, that is a very important gap that deep research doesn't fill but my instrument does.

The instrument

`epistack_eval.py` is a dependency-light script that runs the evaluation. It sends real reasoners, LLM personas, through the knowledge base, and it runs in one command from the linked repository. It is not a mockup with hand-written answers; the reasoning comes from a real model, so the result is not baked in.

The evaluation is one script in two parts:

1. a linkage-integrity test, and
2. a behavioral double-counting test.

Both are designed to measure different failures.

The linkage test measures all three constraints it applies:

1. reasoning only from the base,
2. updating on a held-out claim it never saw, and

3. holding up under a counter-argument.

The first is enforced by instruction. The next two are measured directly. It then also reports the objective metric, calibration gained per claim read.

How it scores

For each persona it records the confidence they end up with, how wide their uncertainty is, whether they were able to identify the load-bearing crux, whether they double-counted correlated clues, and whether their confidence survived the counter-argument. It then compares that against the warranted shape from the public record: the structure of the disagreement, which claims are load-bearing, which clues are correlated, and how wide uncertainty has to be.

For a contested question there is no single true answer to calibrate against, so the instrument does not use one. The target is not a probability. It is the structure of the disagreement, which is a fact about the debate, not a verdict on it.

The three cases

The same instrument runs on three differently-shaped cases, and only the knowledge base and the warranted shape change.

- COVID is contested. Wide uncertainty is warranted.
- Black holes is settled. Confident-safe is warranted, and the failure to catch is false balance.
- Eggs is ill-posed. The failure is a confident universal answer instead of catching that the effect depends on who you are.

The first case, COVID, is a genuinely contested question with an unusually complete public record. Six analysts spanned twenty-three orders of magnitude, neutral analysts differed fiftyfold, superforecasters landed near 75/25 for a natural origin, and the public polls around two-thirds for a lab origin. The facts are largely agreed; the disagreement is driven by the reference-class choice, not the facts. That gap between confident opinion and the careful-but-uncertain read is the kind of thing the instrument is built to detect.

What the evaluation found

Two results held across every run, and were run on both the Sonnet and Opus models. Reasoner personas transferred reasoning to the held-out claim, updating in the correct direction, by a sensible amount, and for a reason that used the claim. Their confidence held under a direct counter-argument, and it did not collapse. Both were measured by the instrument.

The crux localization is model-dependent. For COVID it was patchy on the lighter Sonnet model (0.67, and that too kept moving between different runs) and very reliable on the stronger Opus model (1.0). Held-out transfer and confidence survival were 1.0 across models. This led me to understand that crux localization is fragile and model-dependent, which is exactly what the instrument aims to show.

The double-counting effect was small: 0.05 on the lighter Sonnet, and 0.02, not firing, on Opus. Reasoners moved a little further toward a lab origin when correlated clues were shown separately as compared to when flagged as one, which was both modest and fading on Opus.

Compounding across the three cases

The settled case, black holes, gave the clearest positive result. On the stronger Opus model, calibration gained per claim read was about 0.11, and mean calibration increase about 0.57, with the most skeptical persona gaining most (0.75). Each persona moved to confident-safe, and none fell for false balance when the counter-argument pushed them to doubt a settled answer.

There is another important finding, that the metric is model-dependent. On the contested COVID case it was slightly negative on the lighter Sonnet model (minus 0.01) and slightly positive on the stronger Opus one (0.005). On settled black holes it was clearly positive on both, strongest on the stronger Opus model (0.11). It pointed to the fact that a better reasoner extracts more warranted calibration from the same base.

The eggs case is ill-posed, and it behaved as predicted. On the lighter Sonnet model, crux localization was 0.0: the reasoners missed that the effect is heterogeneous across people, and answered as posed. On the stronger Opus model it was 1.0, meaning they caught it. The metric stayed small and honest either way (0.004 to 0.009).

One caveat: the held-out transfer for black holes scored 0.0 on both the models. The held-out claim reinforced an answer that is already at zero, so there was actually no room left to deviate, and the instrument reported no update as there was no transfer. I note it rather than dress it up.

Since the knowledge base is an input, the instrument can be pointed at any other entrant's knowledge base too, which can make it a shared way to test whether anybody's linkages actually hold.

What it does not show

These are its limitations:

1. These are LLM personas across a few runs, and not a cohort of humans. This means the results are more pointing than proving, with the named models.
2. The key numbers kept moving on different runs and between models, so I report stability, instability, and model-dependence instead of single figures.
3. The black-holes and eggs knowledge bases are compact, defensible starters, meant to show the instrument travels, not authoritative verdicts on those cases.
4. Effort is proxied by claims read, and not real human time. A human cohort, more runs, and larger bases built to stress these paths with more constraints would be my next steps.

Other constraints I did not build

This instrument tested three constraints. I did think of others as well, and they would have strengthened it, but I did not build them, to keep the instrument sharp and shippable. The other constraints that crossed my mind:

1. Order sensitivity: present the same evidence in a different order and see if the conclusion changes. I left it out to keep runs comparable.
2. A planted weak or retracted claim: slip in an authoritative-looking but thin claim and measure whether the reasoner leans on it. This would be the most valuable next addition, because it tests provenance directly, and I skipped it to ensure timely shipping.
3. A distractor: add an irrelevant but plausible claim and see if it moves the needle.
4. Consistency across restarts: run the same persona twice and measure variance, which my results suggest would itself be revealing, since the numbers already move run to run.
5. Authority pressure: tell the reasoner most experts think X and see if it caves.
6. Missing-evidence awareness: ask what it would need to resolve the question, where a good answer names the reference-class gap.

In an ideal world, with more resources, I would pressure-test some of these and add them. I name them deliberately here, because knowing which tests would extend the instrument, or contain it, is my intentional effort to show methodological transparency for this submission.

It matters

This is a working instrument that measures whether a knowledge base's rigor survives a reasoner, on real cases, with reproducible results a reviewer can run. We now have an instrument that can test calibration. That is the difference between an argument about calibration and an instrument that tests it.

Why did I make it

Influencing and leading people from data, to information, to sound decisions under ambiguity and uncertainty at corporate scale is my job in learning design and employee transformation. That is where the instinct comes from, that rigor is only impactful if it survives contact with the recipient. I built this instrument to make that instinct measurable.

Running it

It runs on the following: one file, a command, and an API key. It sends the LLM personas through each case's knowledge base under a knowledge mask. It then applies the three constraints and prints the linkage report, the double-counting result, and the calibration-increase-per-reading-effort metric. The knowledge base and the warranted shape are its inputs, so it runs on other cases unchanged. The run outputs are in the repository, and any reviewer can rerun and compare.

Repository: https://github.com/vaibhav-labs/Epistack_Eval

Appendix: Run Results

Epistack Evaluation: Run Results

Following are the results from running the instrument on the three cases using two models (Sonnet and Opus). It was done using LLM personas, so the results are pointing and not proving.

The main results

Case	Shape	Model	Found the load-bearing crux	Transferred reasoning to held-out claim	Confidence survived counter-argument	Metric (calibration per claim read)
COVID	contested	Sonnet	0.67	1.0	1.0	about -0.01
COVID	contested	Opus	1.0	1.0	1.0	about +0.005
Black holes	settled	Sonnet	1.0	0.0 (see note)	1.0	about +0.09
Black holes	settled	Opus	1.0	0.0 (see note)	1.0	about +0.11
Eggs	ill-posed	Sonnet	0.0	1.0	1.0	about +0.009
Eggs	ill-posed	Opus	1.0	1.0	1.0	about +0.004

What the numbers say

For the settled case of black holes, it gave the clearest positive result (rightfully so). On Opus the mean calibration increase was 0.57, and the most skeptical persona gained the most (0.75). None of the personas fell for false balance when the counter-argument pushed them to doubt a settled answer.

The metric pointed to being model-dependent. It came out cleaner and more positive on the stronger model, and weaker and slightly negative on the lighter one. Which again proved the point that a better reasoner pulls more warranted calibration out of the same base.

Even the crux localization is model-dependent. It was more reliable on Opus and patchy on Sonnet, and it kept moving between different runs.

For the eggs case, it showed its characteristic failure. On the Sonnet model the reasoners missed that the effect is heterogeneous across people, while on Opus they caught it.

The black-holes held-out note

The held-out claim transferred zero on black holes for both models, as it is a settled case. It is a scoring edge. The held-out claim reinforced an answer that was already at zero, so there was no room left to move, and hence the instrument acted correctly in recording no update and no transfer.

Limitations

1. It used LLM personas and not a human cohort, though it had multiple runs.
2. It scored against the structure of each case.
3. The effort metric is proxied by the claims it read.
4. The black-holes and eggs knowledge bases are compact, defensible starters, which are meant to show the instrument travels, and not necessarily authoritative verdicts.